Supplementary information for

Photoluminescence from Cr^{3+} excited orbital states in magnetically ordered $CrCl_3$

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Supplementary Note 1. Calculation details.

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Supplementary Note 2. Transition-dipole analysis of the excited state

The excited-state transition dipole analysis was performed using the VASP WAVECAR obtained from the constrained excited-state calculation at the relaxed excited-state geometry. The relevant hole and electron states were identified from orbital-resolved electronic projections. To locate the dominant contributions to the localized excitation, we evaluated the simultaneous Cr-d localization of the occupied and unoccupied states for all k-points and selected those exhibiting the largest combined localization on the excited Cr ion.

The transition dipole matrix elements were subsequently evaluated using the Python package VaspBandUnfolding and its vaspwfc.py module,[1, 2] which reconstruct pseudo-wavefunctions directly from the VASP WAVECAR file. The calculations were performed for the spin-orbit-coupled noncollinear electronic structure using the corresponding spinor wavefunctions. The dipole matrix elements were obtained through the momentum-matrix formalism implemented in vaspwfc.py, which evaluates the pseudo-wavefunction matrix elements between the selected initial and final states and converts them to dipole-like transition amplitudes through the standard momentum-position relation.

For each selected k-point, the transition dipole vector $d=(d_x,d_y,d_z)$ was evaluated between the occupied and unoccupied excited-state orbitals. The polarization anisotropy was quantified through the corresponding dipole intensities $|d_\alpha|^2$. In the employed lattice convention, the Cartesian x-direction coincides with the crystallographic a-axis.

To visualize the spatial anisotropy of the excitation, charge-density redistribution plots were constructed from partial charge densities corresponding to the excited-state LUMO and ground-state HOMO evaluated at the relaxed excited-state geometry. The resulting real-space charge redistribution was analyzed in the crystallographic ab-plane.

Supplementary Note 3. Polarization during remagnetization

The micromagnetic energy of $CrCl_3$ in an external in-plane magnetic field comprises two terms: the energy of interactions between the magnetization vectors of neighboring monolayers and the energy of interactions between the magnetization vectors of the monolayers and the magnetic field.[19] For a magnetic field $\mu_0 \boldsymbol{H}$ applied along zero-angle,

$$U(\alpha_{1,}\alpha_{2,\dots}H)=\mu_0 M_s H_{ex} \sum_{i,i+1}^{N-1} cos(\alpha_{i+1}-\alpha_i)-\mu_0 M_s H \sum_i^N cos(\alpha_i) \quad (1)$$

where α_i is the angle between the i[th] monolayer's magnetization $\boldsymbol{M_s}$ and the applied magnetic field and H_{ex} is the exchange interaction field (0.1 T for bulk $CrCl_3$).

This equation can be solved easily for a bilayer, as demonstrated in Ref.[3]. When an external magnetic field is applied to a bilayer of $CrCl_3$, the magnetization of the monolayers becomes orthogonal to the external field and the layers remain in antiferromagnetic order. This means that spin flop in the bilayer takes place at 0 T.[3] The magnetization cants towards the external magnetic field as the field is increased until a so-called spin flip occurs, when the magnetizations of both monolayers align with the external magnetic field of H_{ex},[3] as visualized schematically in Fig. S9a. In this case, the PL intensity of a bilayer with $\alpha_1(H=0) = 0$, measured with the analyzer set to $\alpha_a = \alpha_1, \alpha_1 + 90°$, follows the trend shown in Fig. S9b. The polarization direction switches from $90°$ to $0°$ at $H = 0.7\, H_{ex}$. At this field, the polarization degree is also the lowest. This behaviour can be expected by noticing that the polarization of light has a periodicity of $180°$.

There are no unique solutions for more than two layers. For three layers, if one assumes that $\alpha_1 = \alpha_3$ at any field and that $\alpha_{1,3}(H=0) = 0$, the PL intensity in bases set to $\alpha_a = \alpha_1, \alpha_1 + 90°$ follows the trend shown in Fig. S9c. The polarization is constant up to $H = 0.5\, H_{ex}$, when a spin flop takes place. It then starts rotating and the DoP is reduced as a result of a strong increase in the minimum PL intensity polarization component and a weaker maximum reduction. The most significant rotation, of approximately 45°, is observed around $H = 0.8\, H_{ex}$. Spin flip and saturation of polarization occur at $H = 1.5\, H_{ex}$.

These two examples demonstrate that the magnetization process is complex and non-trivial to model, especially for thicker layers, where defects can locally pin the magnetization. However, according to Wang *et al.* [3], in the limit of bulk $CrCl_3$ spin flop should occur as soon as the field is applied ($H = 0$), while spin flip should take place at $H = 2H_{ex}$, in good agreement with our measurements.

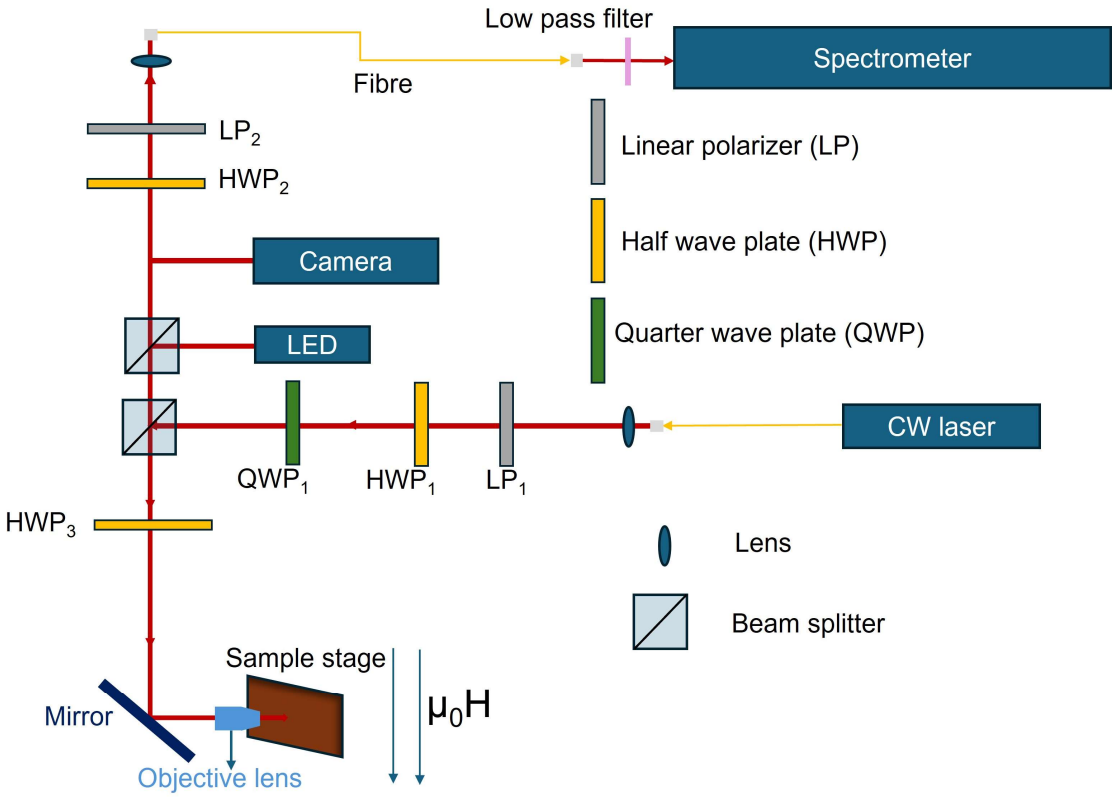


Fig. S1. Sketch of the polarization-resolved photoluminescence measurement setup.

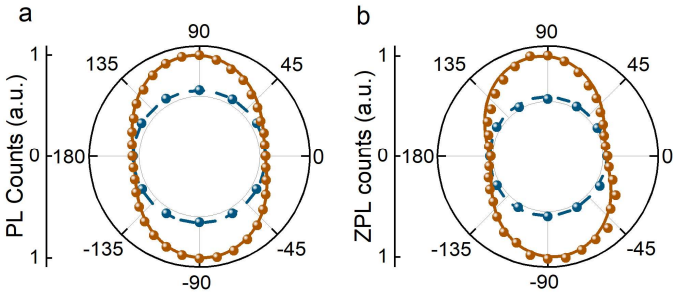


Fig. S2. Integrated PL counts as a function of laser incidence angle. (a) Integrated PL counts (recorded at the same position as that in Fig. 1) plotted as a function of laser incidence polarization angle with the detection analyzer angle fixed at H (0º line), shown using blue symbols. Brown symbols are PL counts of the vibronic signal plotted as a function of detection analyzer angle, with the PL counts normalized to the maximum counts. (b) Equivalent plots for the ZPL signal.

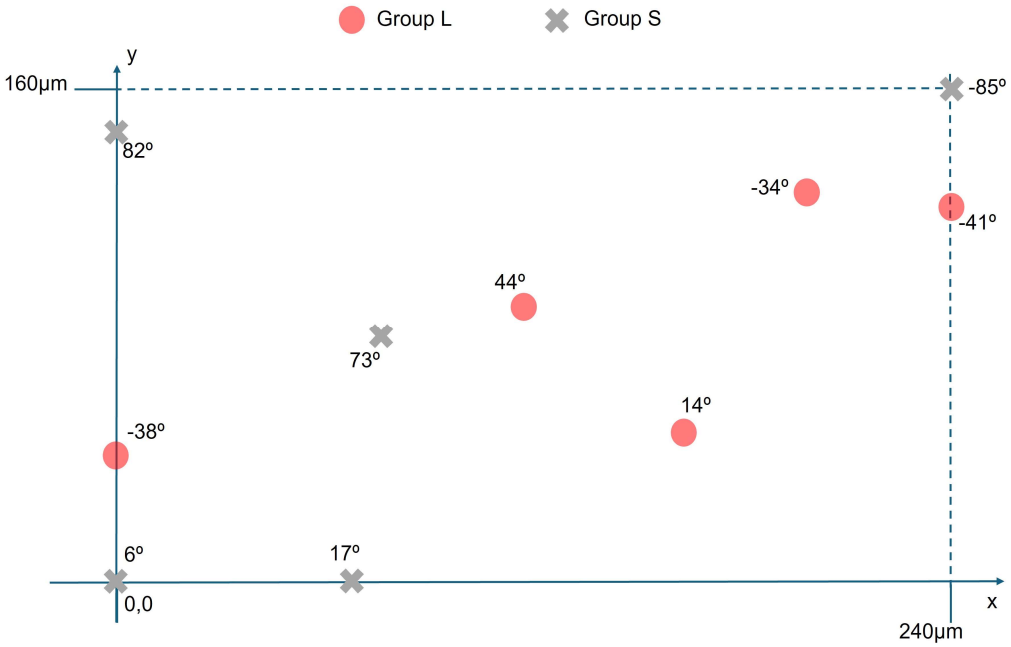


Fig. S3. Relative positions on a rectangular grid of 240 μm by 160 μm. The points are chosen with clear ZPL signals. Each position is marked with its polarization angle at zero magnetic field, along with a retrospectively-assigned Group (L or S), See Fig. 4. The colors are chosen to match those in Fig. 4.

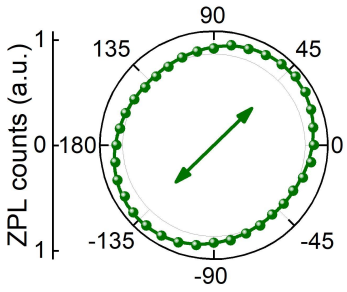


Fig. S4. PL counts of ZPL plotted as a function of analyzer angle at 2 K. The arrow indicates the polarization angle at 44º.

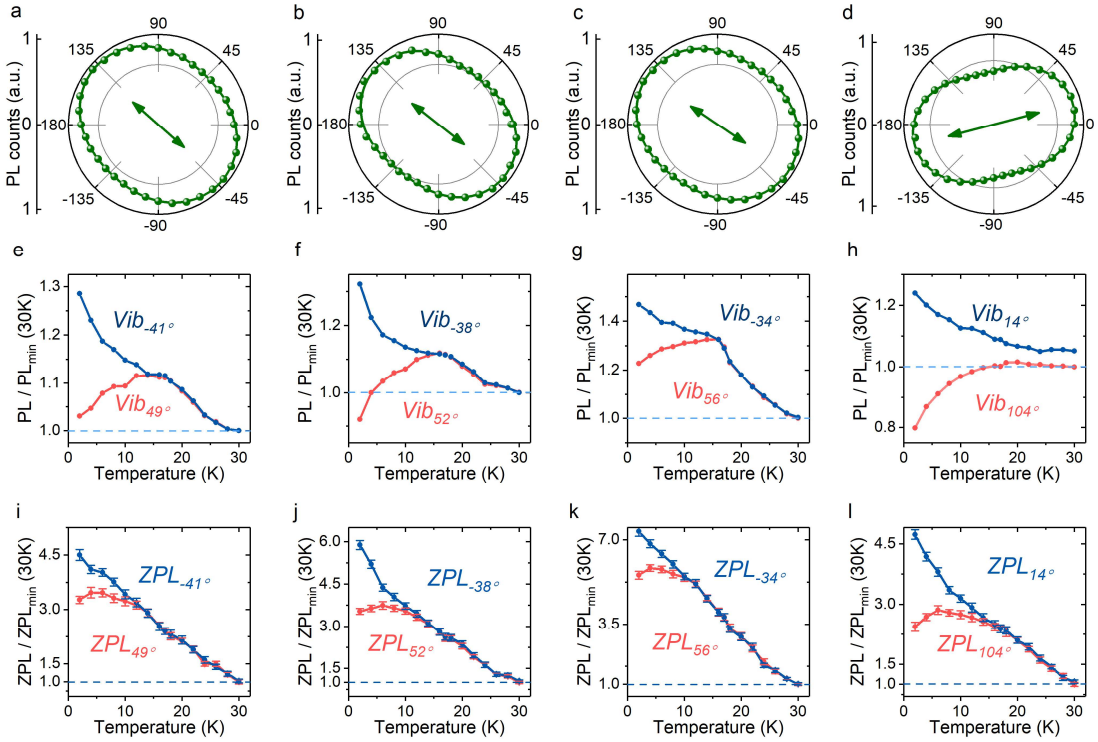


Fig. S5. Polarization-dependent PL emission recorded from positions in Group L. (a-d) Polar plots of PL counts plotted as a function of polarization detection analyzer angle at 2 K, with the radial scales normalized to the maximum value in each plot. The arrows show the preferred polarization angle. The polarization angles are -41º, -38º, -34º and 14º, respectively. (e-h) Temperature-dependent evolution of vibronic PL counts that along the preferred polarization direction and its orthogonal counterpart, normalized to the minimal PL counts at 30 K, corresponding to the points in (a-d). (i-l) Temperature-dependent evolution of the ZPL signal that along the preferred polarization direction and its orthogonal counterpart, normalized by the ZPL count minimum value at 30 K.

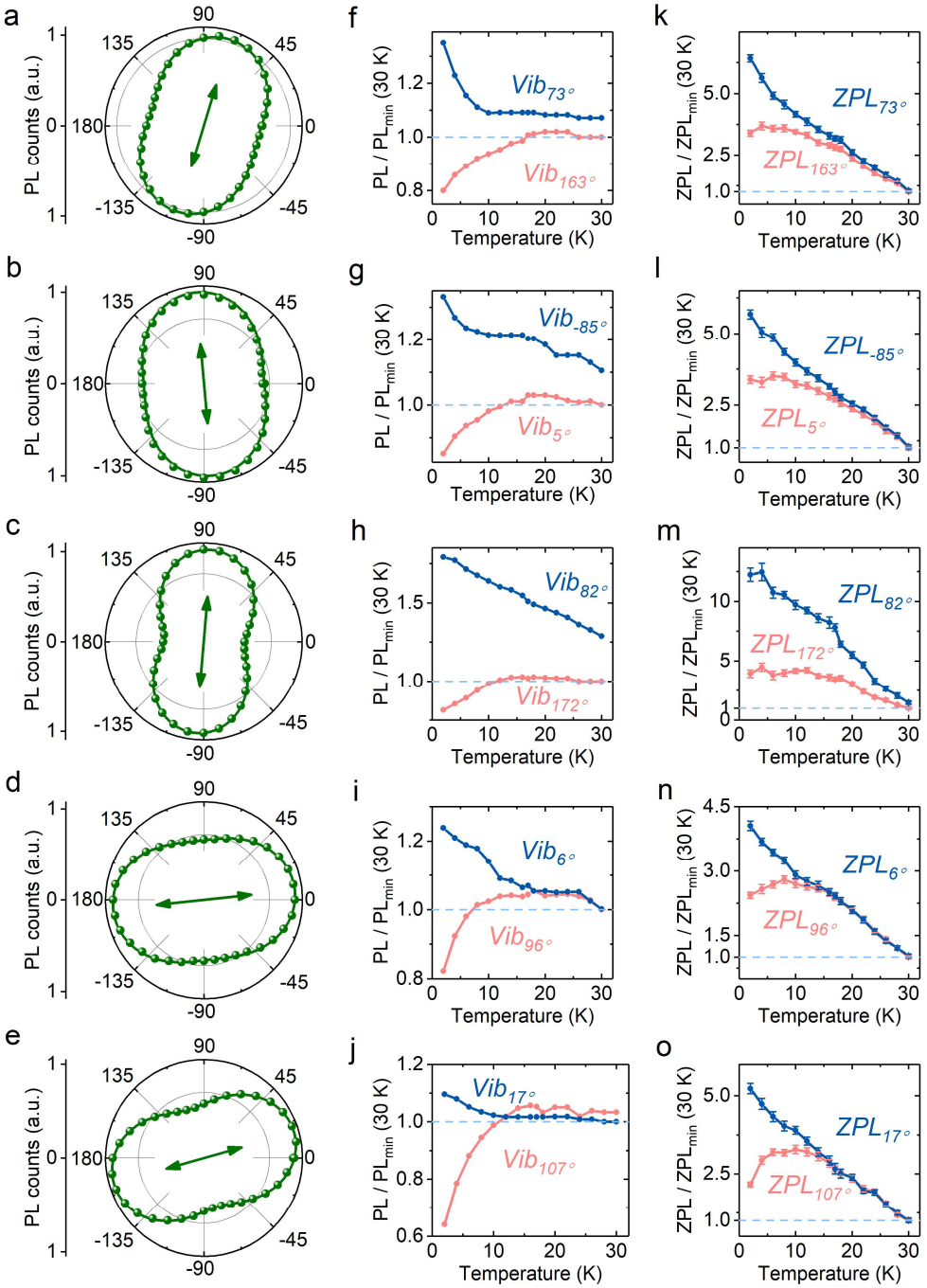


Fig. S6. Polarization-dependent PL emission recorded from points in Group S. (a-e) Polar plots of PL counts plotted as a function of polarization detection analyzer angle at 2 K, with the radial scales normalized to the maximum value in each plot. The arrows show the preferred polarization angle. The polarization angles are 73º, -85º, 82º, 6º and 17º, respectively. (f-j) Temperature-dependent evolution of vibronic PL counts along preferred polarization direction and its orthogonal counterpart, normalized to the minimal PL counts at 30 K, corresponding to the points in (a-e). (k-o) Temperature-dependent evolution of the ZPL signal along the preferred polarization direction and its orthogonal counterpart, normalized by the ZPL count minimum at 30 K.

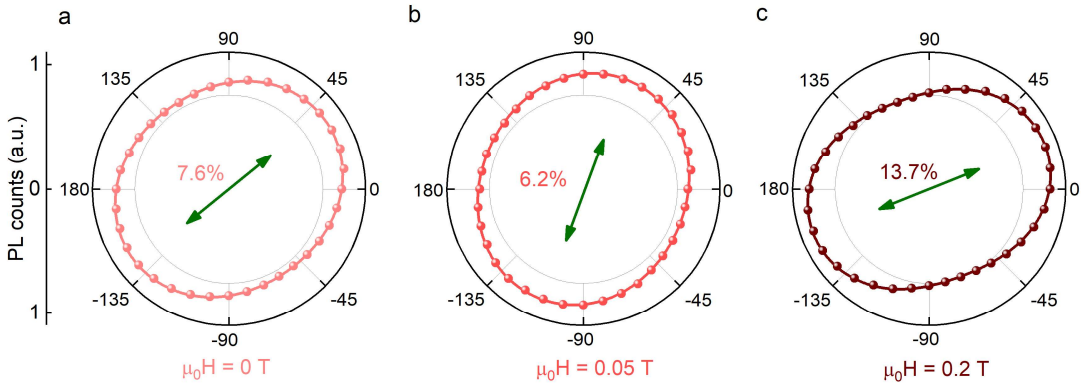


Fig. S7. Evolution of integrated PL counts of vibronic signal with magnetic fields for the same location as in Fig. 3. (a-c) Polar plots of vibronic signal as a function of detection analyzer angle with magnetic fields at 0, 0.05, and 0.2 T, respectively, shown on the same radial scale, normalized to the maximum. Solid lines show fitted functions. Green arrows mark the preferred polarization angle.

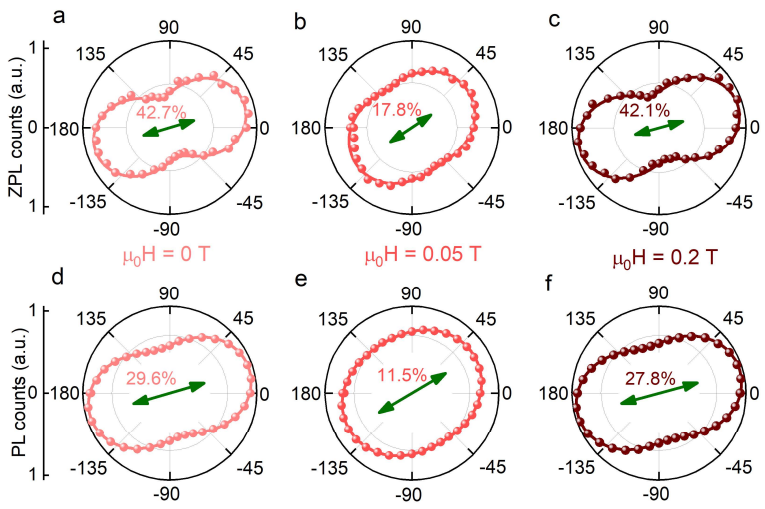


Fig. S8. Evolution of polarization-dependent PL emission with magnetic field at one location in Group S, with polarization angle 17° in zero magnetic field. (a-c) Polar plots of the ZPL signal plotted as a function of detection analyzer angle in external magnetic fields of 0, 0.05 and 0.2 T. The plots have the same radial scales and are normalized to the maximum counts. (d-f) Equivalent plots for the vibronic signal. The solid lines in (a-c) and (d-f) are fitted curves. Green arrows mark the preferred polarization direction. The DoP is given above the arrow.

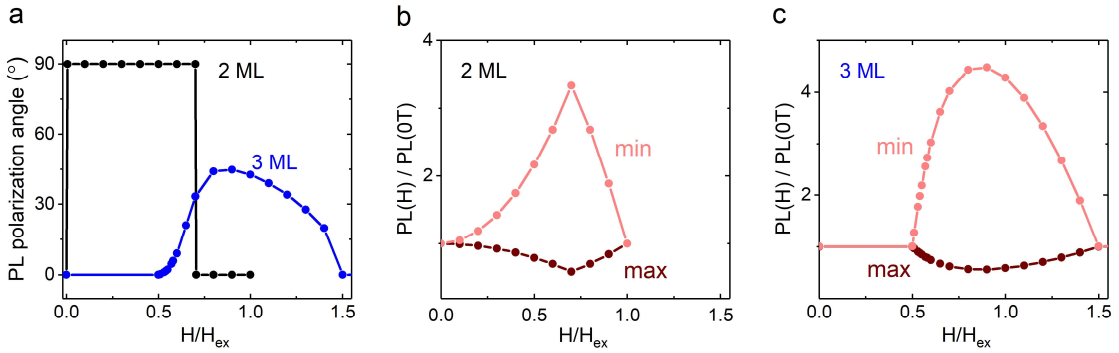


Fig. S9. Simulations of polarization angles with different stacking. (a) Direction of polarization plotted as a function of external magnetic field for a bi-layer (2 ML) and a tri-layer (3 ML) $CrCl_3$. H_{ex} is the exchange interaction field. The polarization at 0 T and the external field are co-linear in this example. (b) Evolution of PL intensity for a 2 ML film along 0º and 90º as a function of magnetic field, labeled min and max. (The polarization direction at 0 T is 0º). Both components are normalized to their values at 0 T. (c) Evolution of PL intensities for a 3 ML film along 0º and 90º, plotted as a function of magnetic field, labeled min and max. (The polarization direction at 0 T is 0º). Both components are normalized to their values at 0 T.

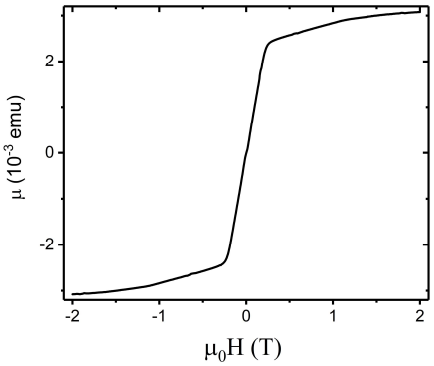


Fig. S10. Magnetization curve of $CrCl_3$ plotted as a function of in-plane magnetic field. The field is swept from -2 to 2 T. The reverse sweep (showing minimal hysteresis) is not shown for clarity.

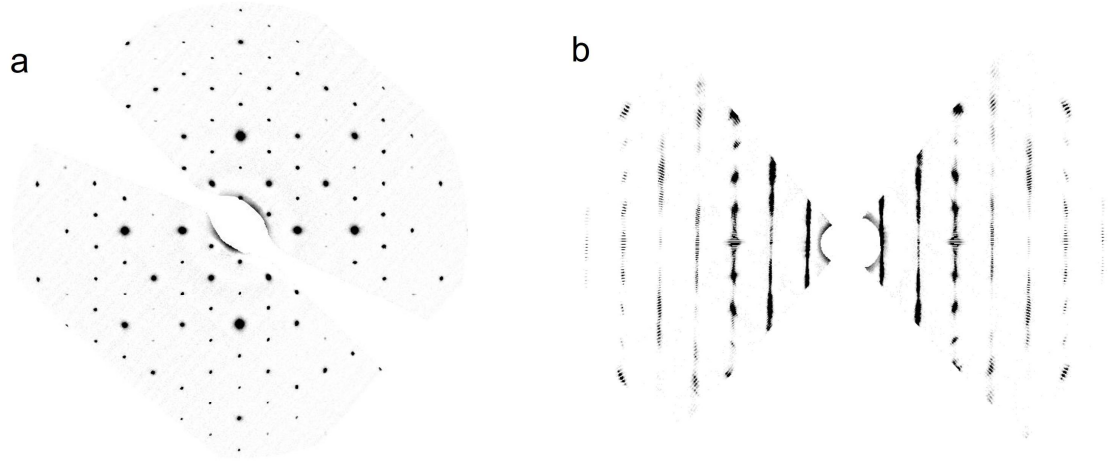


Fig. S11. Crystal structure characterization of $CrCl_3$ by electron diffraction. (a) Electron diffraction spectra recorded from exfoliated bulk-like $CrCl_3$, showing a hexagonal metric. The crystal was oriented with the [001] direction along the incident electron beam direction. (b) Lines of diffuse intensity, characteristic of stacking faults, were observed in a [100] zone-axis diffraction pattern.

Supplementary References:

1. Q. Zheng. VaspBandUnfolding. GitHub repository, https://github.com/QijingZheng/VaspBandUnfolding.
2. Q. Zheng. Light–Matter Interaction and Dipole Transition Matrix. https://qijingzheng.github.io/posts/Light-Matter-Interaction-and-Dipole-Transition-Matrix/.
3. Z. Wang, M. Gibertini, D. Dumcenco, T. Taniguchi, K. Watanabe, E. Giannini and A.F. Morpurgo. Determining the phase diagram of atomically thin layered antiferromagnet $CrCl_3$. Nat. Nanotechnol. 2019, **14** (12), 1116-1122.